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La Liga

Data Visualization

Introduction

This project presents an interactive dashboard visualizing two decades of Spain's La Liga football data. The goal is to allow fans and analysts explore various aspects of the league's history, from championship races in individual seasons to long-term team performance trends. We also highlight the era dominated by two legendary players, Lionel Messi and Cristiano Ronaldo, through a dedicated interactive feature. Developed with the library D3.js, the dashboard integrates five coordinated visualizations, enabling users to investigate different facets of the game (standings, match outcomes, discipline, and scoring efficiency) in a unified interface. The emphasis is on clarity and engagement, providing intuitive visuals and interactions that help uncover patterns and stories in the data.

The motivation for this visualization stems from the rich narrative of La Liga between 2005 and 2025. This period includes intense title rivalries (especially between Barcelona and Real Madrid), the rise and fall of various clubs, and the extraordinary performance of Messi and Ronaldo. By visualizing the data, we aim to illustrate these stories: how the championship battles unfold each year, how teams fare against each other, which teams play more aggressively or efficiently, and how two superstar players' careers align with league trends. The project focuses on delivering these insights in an interactive way, where users can filter and highlight the information of interest. In the following sections, we describe the dataset and each visualization component, along with the interactive features and design decisions that shaped the final dashboard.

I. Data Description and Processing

The dashboard relies on historical match data from the 2005/06 season up to 2024/25. The data was compiled from season archives and includes every match's key statistics: the teams involved, goals scored, shots taken, fouls committed, yellow/red cards, and the match outcome (win, loss, or draw). Each season's data is stored in a CSV file, and we load all seasons to enable both season-specific and longitudinal visualizations. Before visualization, we perform some processing to derive summary metrics needed for certain charts. For example, for each season we calculate the final standings (rankings) of teams based on total points (with goal difference and goals scored as tiebreakers), since the raw match data needs to be aggregated to determine each team's rank. We also compute team-level aggregates per season, such as total fouls, total cards, and goals conceded, to feed into the disciplinary chart, and average goals per match or shots per match for the efficiency chart. This data processing is done in JavaScript when the application loads, ensuring that all visualizations are based on consistent and up-to-date computations directly derived from the input data.

One challenge was ensuring that data from different seasons could be compared or combined. Teams change over the years (some are relegated or promoted), so our longitudinal visualization needed to handle teams appearing or disappearing across seasons. We addressed this by including all teams in a master list and marking missing seasons appropriately. Additionally, because the dataset spans 20 seasons, we implemented a mechanism to navigate through time: a global season slider allows users to select a specific season, updating the season-specific charts accordingly, this slider is visible in *Figure 1*. The data for the selected season is then used to render the yearly visualizations, while the all-time chart always shows the full range but can highlight the selected season or range of interest. With the data prepared, we developed multiple coordinated visual views, each focusing on a different perspective of the league.

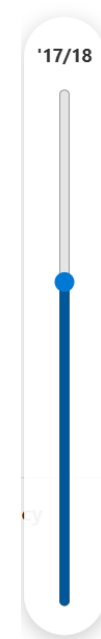


Figure 1 :
Season slider,
its opacity
increases
on hover

II. Visualization Components and Interactive Features

The dashboard is organized into five main visualization components, each in its own panel but linked together through interactive filtering and highlighting. Users can interact with one chart (for instance, selecting a team or a player's era) and see related effects across the others. Below we describe each component in detail and explain the interactions enabled.

Title Race – Season Progression Line Chart

The first panel, titled “The Title Race”, visualizes the progression of team standings over the course of a selected season. It shows how the championship battle unfolds from the first matchday to the last. We represent this with a line chart where the x-axis is the matchday (from 1 to 38 rounds in a typical season) and the y-axis is the cumulative points. Each team is drawn as a line that rises as the team accumulates points week by week. This allows users to see which team led the table at various stages and how close the top contenders were. For example, one can observe if a team pulled ahead early or if there was a dramatic come-from-behind surge late in the season.

All teams are included in the chart, but to prevent it from becoming visually overwhelming, interactive highlighting is used: by default, each line is fairly thin and semi-transparent, and if the user selects one or more teams, those teams’ lines become bold and opaque while the others fade into the background. This selection can be done by clicking on a line directly in the chart (or through other charts or a team list), leveraging a brushing-and-linking approach across the dashboard.

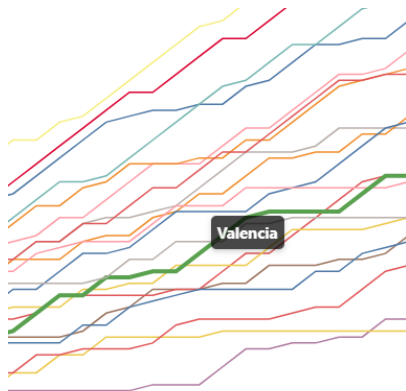


Figure 2 : The Title Race Chart, Valencia is being hovered

We also implemented an animation for this chart: when a new season is selected via the slider, the lines draw themselves progressively from matchday 1 to 38, giving a sense of the race “playing out” over time. Once the lines finish drawing, a subtle animation of dots can be triggered to emphasize the final positions (this was a design choice to connect the outcome of the title race to the overall rankings chart described below). Tooltips are available on hover, displaying the team name and increasing the thickness of the hovered line, this is shown on *Figure 2*, and clicking a team locks the highlight. Overall, the Title Race view provides a dynamic way to relive each season’s competition trajectory.

Who Beat Whom – Match Outcome Network

The second panel, “Who Beat Whom?”, is a circular network diagram that illustrates all head-to-head results between teams in the selected season. In this visualization, each team is represented by a segment (arc) on the circumference of a circle. Every match is shown as a curved link connecting the two teams that played. The link’s color and direction encode the result: a gradient from green to red indicates a home win (green at the home team’s side transitioning to red at the away side), while red-to-green indicates an away win (red at the home side, green at the away side). Drawn matches are represented with a neutral grey link. This design allows users to quickly see patterns such as which team defeated many others (they will have many outgoing green-to-red links) or if there were any teams that two specific teams exchanged wins (both directions colored), it is also a very efficient way to see which teams managed to beat the best teams as they only have a few red links going out of their nodes. The circular layout and colored links give a compact overview of all matches in the season.

Interactivity plays a key role in making this complex network readable. When the user hovers over a team’s arc, a tooltip shows the team name, and all links involving that team are

highlighted while other links are subdued. This way, one can focus on the performance of a single team or of several selected teams as shown in the *Figure 3* — for instance, hovering over a mid-table team will highlight all matches they played, revealing which big teams they might have beaten or lost to. Clicking on a team's arc will select that team, which not only keeps its matches highlighted in this chart but also triggers highlighting of the same team across the other charts (for example, its line in the Title Race chart and its points in other charts will stand out). The Who Beat Whom view is especially useful to spot surprising results or the overall dominance of top teams in a season. We included a slight animation delay when drawing the links, so they appear gradually, which adds a visual rhythm and helps the user digest the information step by step. This visualization was one of the more challenging to implement due to positioning all teams in a circle and drawing many curved links, but it effectively conveys the connectivity of results in a season.

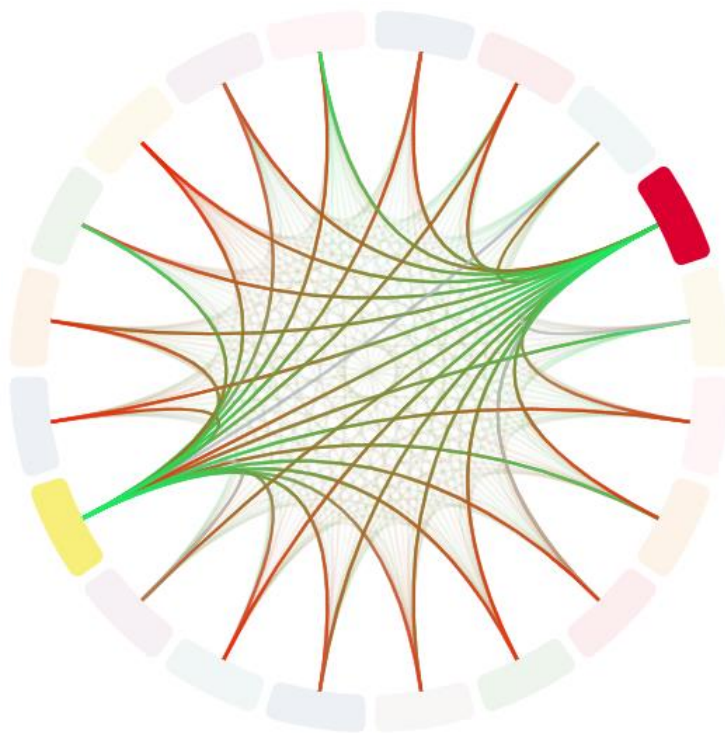


Figure 3 : Who Beat Whom Network, here Barcelona (red) and Real Madrid (yellow) have been selected, only the games concerning one of these two teams are highlighted.

Overall Leaderboard – Historical Bump Chart

The centerpiece of the dashboard is the “Overall Leaderboard” bump chart, which spans all seasons and shows how each team's final ranking changed over the years. Each team is represented by a line that runs horizontally across the chart, with the x-axis representing seasons (from 2005/06 through 2024/25) and the y-axis representing the final rank (1st place at the top down to 20th at the bottom, corresponding to the league positions). This chart provides a long-term view: for example, one can trace the performance of a club like Atlético Madrid or Sevilla over time, see periods of dominance by Barcelona and Real Madrid at the very top, or identify when a team was relegated (their line would drop off after a certain year since they no longer appear in subsequent seasons). We handle teams entering or leaving the league by letting their line become dashed or simply not plotting beyond the seasons they

participated; teams that are not in a given season are essentially missing data for that year, which we leave blank in their line.

The bump chart is dense with information, so we again rely on interactive highlighting to aid readability. Without any selection, all teams' lines are shown in a subdued manner. If a user selects one or multiple teams (either by clicking directly on a line or by using the team selection list provided), the chosen teams' lines are emphasized (drawn with full opacity and thicker stroke) while all other lines fade back. This lets the user isolate the trajectories of particular teams of interest. For instance, comparing Barcelona's and Real Madrid's lines clearly shows their constant presence at the top ranks, whereas a team like Recreativo has a line that disappears after 2009 (indicating relegation).

A special interactive feature in this chart is the "Focus on a Legacy" highlight for Messi and Ronaldo's era. On the right side of the dashboard, we provide two clickable portraits, as shown in *figure 4* – one for Messi and one for Ronaldo. When the user clicks Messi's image, the bump chart responds by coloring the background of the seasons from 2005/06 through 2020/21, the period during which Lionel Messi was active in La Liga. Similarly, clicking Ronaldo's image highlights the seasons 2009/10 through 2017/18, corresponding to Cristiano Ronaldo's time at Real Madrid. We implemented this by overlaying a semi-transparent colored rectangle on the bump chart spanning the relevant seasons. This visual emphasis on the timeline helps contextualize those players' careers relative to the league's history – for example, users can see at a glance which teams were winning titles during Messi's era (mostly Barcelona and Real Madrid, as the chart shows) or how the competitive landscape looked when Ronaldo was present.



Figure 4 : Focus on a Legacy, Messi is selected : overlaying Messi's era in la Liga on the Overall Leaderboard Chart

Originally, we had considered highlighting these players' impact in a more complex way (such as filtering the chart or annotating specific data points), but that approach proved difficult to implement. The current solution of coloring the period of their play was a simpler and effective alternative that still conveys the intended insight. The legacy focus can be toggled on and off by clicking the portraits again, and it works in tandem with team selections – for example, one can select Barcelona and then toggle Messi's era to see how the team performed specifically in those highlighted seasons.

Notably, the bump chart is updated dynamically with the season slider as well. If a user drags the slider to an earlier season, we use that context to only display lines up to that season, thus having access to the full graph is only possible in the last season (2024-2025). When a new season is selected on the season slider, new points appear in the Overall Leaderboard chart, these points come from the Title Race chart with an animation shown in Figure 5.

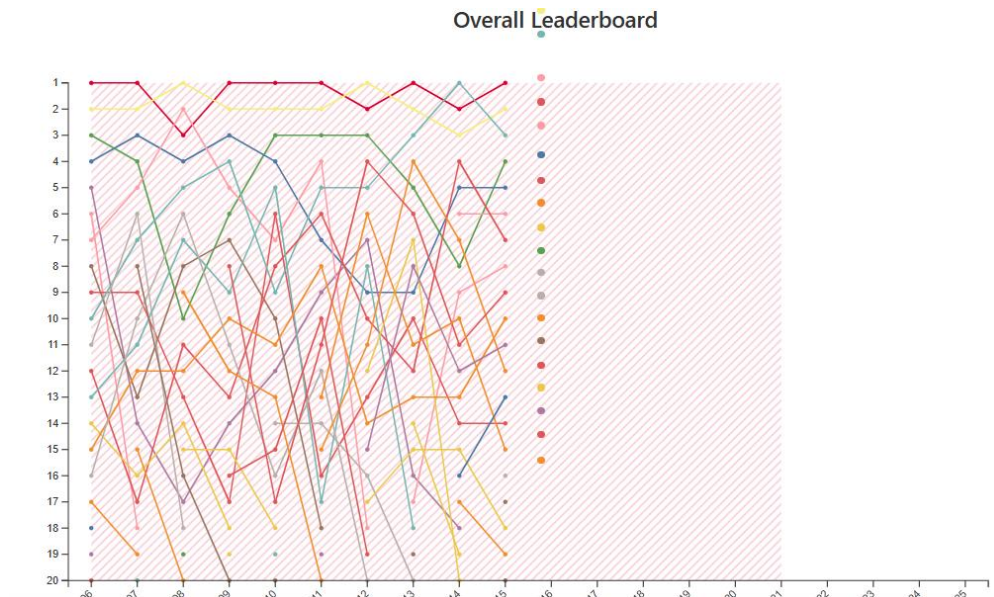


Figure 5 : Overall Leaderboard Chart, Messi is selected and the new points for season 2015/2016 are coming from the Title Race chart.

However, the slider's main effect is on other charts; the bump chart is mostly static aside from the legacy highlight and team filtering. Tooltips on this chart can show the team name when hovering over a point or segment of a line and this hovering also increases the thickness of the hovered line, aiding identification in the dense mesh of lines. The Overall Leaderboard serves as a historical reference that ties together all the season-by-season stories into a big picture, and it remains visible as a backdrop even while the user explores individual seasons on the other panels.

Fouls and Aggressivity – Fouls vs Cards Scatterplot

Another aspect we wanted to explore is the discipline and aggressivity of teams, visualized in the "Fouls & Aggressivity" panel. This view is a scatterplot that compares the total number of fouls committed by each team to the total number of cards (yellow and red combined) they received in the selected season. Each team appears as a single circle on this chart. The x-axis represents the total cards and the y-axis represents the number of fouls. We expect these to be correlated – generally, teams that commit more fouls will accumulate more cards – but the interesting part is seeing which teams are outliers. For example, a team might commit a high number of fouls but receive relatively fewer cards (perhaps indicating lenient refereeing or tactical fouling that avoids bookings), whereas another team could have disproportionately many cards for the fouls they commit (suggesting a more reckless style or stricter officiating on their games).

In our implementation, we added a third dimension to this chart by encoding each team's defensive solidity as the size of the circle: the radius of each circle is inversely proportional to the number of goals the team conceded that season. This means teams with larger bubbles conceded fewer goals (often the stronger teams with good defenses), and teams with very small bubbles let in many goals. This way, one can spot if a very aggressive team (high fouls, high cards) also had a strong defense or not. For instance, a large circle far to the top-right

would indicate a top defensive team that also played quite physically (many fouls and many cards). The color of each circle corresponds to the team's colors for easy identification, but we also provide interactive aids: hovering over a circle will display a tooltip with the team's name and their exact stats (fouls, cards, goals conceded) as shown on *Figure 6*, and as with other charts, clicking a circle will select that team globally.

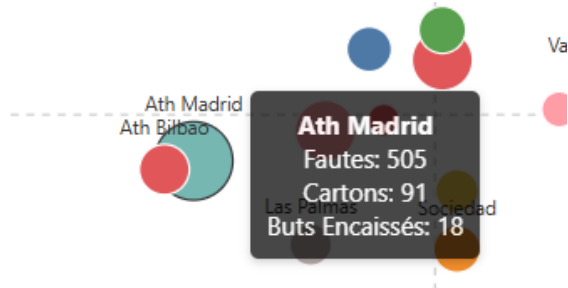


Figure 6 : tooltip from the Fouls and Aggressivity Scatterplot, when hovering over Ath Madrid circle.

Selection causes that team's point to remain highlighted (and it will also highlight in other charts like the bump chart and title race), while non-selected teams in the scatter will become semi-transparent. This linked selection helps in investigating correlations; for example, a user might click on a specific team in this Fouls vs Cards chart to highlight it, then switch the season to see if that team's approach was consistent over years (each season selection will update the scatterplot accordingly). The chart also draws reference lines for the median number of fouls and cards across all teams, effectively dividing the plot into quadrants. This gives a quick sense of whether a team is above or below average on these metrics. Overall, the Fouls & Aggressivity scatterplot adds a behavioral perspective to the dashboard, complementing the performance-focused charts with insight into each team's style of play in terms of physicality and discipline.

Goal Efficiency – Shots vs Goals Bubble Chart

The final panel, titled "Goal Efficiency", examines how effective teams are at converting opportunities into results. It's a scatterplot that plots the average number of shots per match (on the x-axis) against the average number of goals per match (on the y-axis) for each team in the selected season. Essentially, this chart asks: given how many shots a team typically produces, how many goals do they score? Teams that appear higher in the chart manage to score more goals per game, and teams further right take more shots per game. A team in the upper-right corner would be an offensive powerhouse (many shots, many goals), whereas a team in the lower-right might be wasteful (many shots but relatively few goals), and a team in the upper-left could be very efficient (not many shots but a high goal yield). To add more context, we represent each team's total points in the season by the size of the bubble – teams with larger bubbles had more successful seasons in terms of league points. This often correlates with goal-scoring effectiveness, but not always, and the chart makes those differences visible. For example, a mid-table team might have a similar goals-per-match and shots-per-match as a top team, but a much smaller bubble indicating far fewer points (perhaps due to defensive issues or inconsistency).

We divided the Goal Efficiency plot into quadrants with faint guide lines representing the league averages of shots and goals. This helps categorize teams broadly: teams above the horizontal average line score above-average goals per game (good attack), those below score less than

average; teams to the right of the vertical line take above-average shots (volume shooting teams), and those to the left take fewer shots than average. During development, we even labeled the quadrants with descriptive nicknames (such as “Clinical Finishers” for high efficiency or “Profligate Shooters” for teams that shoot a lot with little end product) to make it more readable, but in the final version we kept it clean with just the axes labels and subtle hints, to avoid cluttering the visual. Interaction works similarly to the fouls/cards chart: you can hover to see exact values and team names, and click to select a team. Because this chart shares the same selection mechanism, one neat effect is that if you select a team here, you can immediately see where that team lies in the fouls/cards chart (the other scatter) as well as trace them in the bump chart and title race.

Regardless of the visualization being explored, selecting a team—by clicking on its circle in either scatterplot, its node in the network graph, or its trajectory in one of the two bump charts—automatically selects that team across all other visualizations, as well as in the menu located to the left of the central chart. This menu lists all teams in alphabetical order and allows users to select or deselect any team of interest. It provides an exhaustive list of all teams present in the dataset and also acts as a global legend by displaying the color associated with each team. This menu is shown in *Figure 7*.

Figure 7 : Team selection menu, Elche and Malaga are being selected



This cross-linking encourages comparative exploration – for instance, one might find that a team with very high shooting efficiency one year also was a top performer, or check if that efficiency changed the next year by dragging the season slider. The Goal Efficiency view thus sheds light on the offensive prowess of teams and, combined with the other views, completes the multifaceted analysis of team performance.

Conclusion

In conclusion, our La Liga visualization dashboard brings together multiple interactive charts to provide a comprehensive view of the league's dynamics from 2005 to 2025. By coordinating these different visualizations – from the temporal drama of each season's title race to the overall historical context and specific statistical insights – users can explore the data at both granular and summary levels. The interactive features such as team selection, season slider, and the Messi/Ronaldo legacy highlight help users tailor the experience to their interests, whether they want to follow a single team's journey or compare playing styles across teams and eras. This focus on certain data features is possible thanks to the context/focus display techniques seen in class, such as playing on opacity and overlaying new information to focus on certain features.

Through this project, we have illustrated key stories in the data: the dominance of certain clubs, the impact of legendary players, and the varying strategies of teams in terms of offense and discipline. The design and implementation involved simplifying complex ideas (like highlighting a player's era) into intuitive visual cues, and ensuring the dashboard remains responsive and clear despite the breadth of data. There were trade-offs and adjustments during development – for example, opting to color part of the timeline for the legacy focus instead of a more complicated filtering mechanism – all aimed at improving usability and clarity.

The final outcome is an exploratory tool that provides genuine insights into two decades of Spanish football. If given more time or resources, future enhancements could include adding more player-specific data (to complement the Messi/Ronaldo focus with other stars), incorporating other user queries (like highlighting all championship-winning teams), but also using other features from our dataset, like data on betting odds for example.